GIT QUEST | 30-DAY GIT LEARNING SERIES

Day 5: git log, Your Time Machine

Week 1: First Steps | Level: Absolute Beginner

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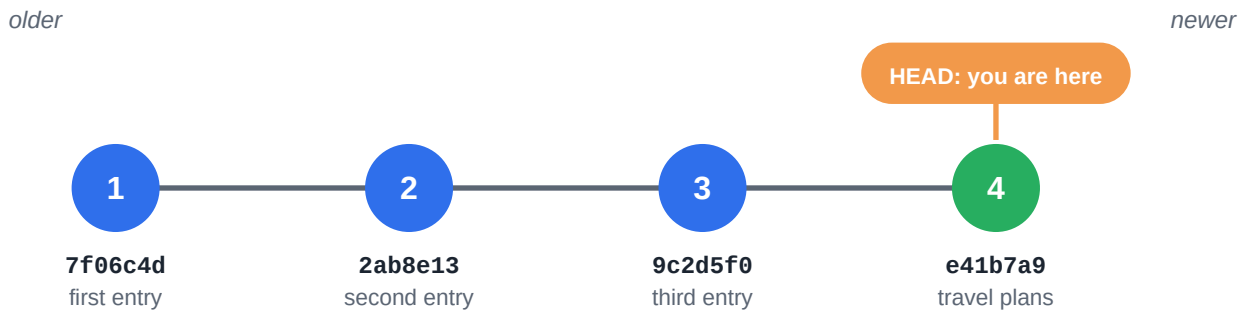
Today's mission

Yesterday you learned to write clear labels for your save points. Today you learn to **travel back and read them**. `git log` is the time machine of Git.

- Read the full output of `git log`
- Zoom in and out with `--oneline`, `-n`, `--stat` and `--graph`
- Search your history and open one commit with `git show`
- Escape the log screen without panicking

1. The big idea: the time machine

Every commit is a save point on a timeline. `git log` lists them all, **newest first**. The newest one is marked **HEAD**, a pin that says "you are here".



2. Reading one log entry

```
commit e41b7a9c3d1f2a8b6e0c5d4f7a9b1c2d3e4f5a6b (HEAD -> main)
Author: Your Name <you@example.com>
Date:   Mon Nov 9 10:42:17 2026 +0530

    Add travel plans
```

Part	What it means
e41b7a9c3d1f...	The commit's unique ID, called a hash. The first 7 characters are enough
(HEAD -> main)	You are here: the commit you are on, on branch main
Author and Date	Who saved it (from git config on Day 1), and when
Add travel plans	The label you wrote. This is why Day 4 matters

3. Build a tiny history

```
git init -b main time-machine
cd time-machine

echo "Went to the market" > diary.txt
git add diary.txt
git commit -m "Add first diary entry"

echo "Learned Git today" >> diary.txt
git add diary.txt
git commit -m "Add second diary entry"

echo "Baked a cake" >> diary.txt
git add diary.txt
git commit -m "Add third diary entry"

echo "Planning a trip" > plans.txt
git add plans.txt
git commit -m "Add travel plans"
```

4. Five zoom levels

Zoom 1: everything (press q to leave a long list)

```
$ git log
```

Zoom 2: one line each

```
$ git log --oneline
e41b7a9 (HEAD -> main) Add travel plans
9c2d5f0 Add third diary entry
2ab8e13 Add second diary entry
7f06c4d Add first diary entry
```

Zoom 3: only the last few

```
$ git log --oneline -n 2
e41b7a9 (HEAD -> main) Add travel plans
9c2d5f0 Add third diary entry
```

Zoom 4: which files changed?

```
$ git log --stat -2
 e41b7a9 Add travel plans
 plans.txt | 1 +
 9c2d5f0 Add third diary entry
 diary.txt | 1 +
```

Zoom 5: the map

```
$ git log --graph --oneline
* e41b7a9 (HEAD -> main) Add travel plans
* 9c2d5f0 Add third diary entry
* 2ab8e13 Add second diary entry
* 7f06c4d Add first diary entry
```

5. Search and open

```
$ git log --oneline --grep="diary"
9c2d5f0 Add third diary entry
2ab8e13 Add second diary entry
7f06c4d Add first diary entry
```

```
$ git show 9c2d5f0
commit 9c2d5f0...
    Add third diary entry

@@ -1,2 +1,3 @@
 Went to the market
 Learned Git today
+Baked a cake
```

The line starting with + is what that commit added. We will decode it fully on Day 6.

Common problems

You see	What to do
Screen stuck with : or (END)	Git opened a pager. Press q to quit. Space = next page, arrow keys = scroll
does not have any commits yet	Make a commit first
fatal: ambiguous argument	The hash is wrong. Copy the 7 characters from git log --oneline

6. Cheat sheet

Command	What it does
git log	Full history, newest first
git log --oneline	One line per commit
git log --oneline -n 3	Only the last 3 commits
git log --stat	Shows which files each commit touched
git log --graph --oneline	Draws the history as a map
git log --grep="word"	Finds commits whose message has a word
git show <hash>	Opens one commit in detail

Bonus: make your own shortcut

```
git config --global alias.lg "log --oneline --graph --decorate"
git lg
```

7. Today's 3 commits

#	File	Content	Commit message
1	day05-git-log/lesson.md	Today's lesson with zoom levels	docs(day05): add lesson on git log time machine
2	day05-git-log/demo.tape	VHS script that records the terminal GIF	feat(day05): add VHS tape for git log demo
3	exercise.md and days-1-to-5-summary.md	Challenge, plus a recap of Days 1 to 5	docs(day05): add exercise and days 1-5 summary

8. Mermaid diagram GitHub draws for you

````
```mermaid
gitGraph
  commit id: "Add first diary entry"
  commit id: "Add second diary entry"
  commit id: "Add third diary entry"
  commit id: "Add travel plans"
```
````

9. Challenge and quiz

Challenge: use `git log --stat` to find the commit that added `plans.txt`, then open it with `git show` and its 7-character ID.

Q1. Which command shows one line per commit?

Q2. What does HEAD mean?

Q3. How do you leave the log screen when it opens?

Answers: 1) git log --oneline. 2) The commit you are on: "you are here". 3) Press q.

> **Coming up: Day 6**
> **git diff, spot the difference.** See exactly which lines changed between two versions.